

Primary Result

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1 Gabber's observation

Occasionally, I read a very interesting result which Gabber mentioned in [1, Lemma A5]. I quote it here with proof following the sketch given by Peter Scholze at the website [2].

Lemma 1. *Let K be a henselian valuation field with residue field k , K^s and k^s the separable closure of K and k respectively. Then the canonical exact sequence*

$$0 \longrightarrow I_K \longrightarrow \mathrm{Gal}_K \longrightarrow \mathrm{Gal}_k \longrightarrow 0 \quad (1)$$

splits, where I_K is the inertia group of K^s/K , $\mathrm{Gal}_K := \mathrm{Gal}(K^s/K) \cong D(K^s/K)$ the decomposition group and $\mathrm{Gal}_k := \mathrm{Gal}(k^s/k)$.

Proof. First to reduce to K' with maximal property in the sense that there is no algebraic extension K''/K' whose residue field k'' is purely inseparable over k' . For any algebraic extension K'/K with k'/k purely inseparable, suppose that the following exact sequence splits

$$0 \longrightarrow I_{K'} \longrightarrow \mathrm{Gal}_{K'} \longrightarrow \mathrm{Gal}_{k'} \longrightarrow 0 \quad (2)$$

then there is a section $s' : \mathrm{Gal}_{k'} \rightarrow \mathrm{Gal}_{K'}$. Consider the composition $s : \mathrm{Gal}_k \xrightarrow{\sim} \mathrm{Gal}_{k'} \xrightarrow{s'} \mathrm{Gal}_{K'} \rightarrow \mathrm{Gal}_K$. Since there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Gal}_{K'} & \longrightarrow & \mathrm{Gal}_{k'} \\ \downarrow & & \downarrow \sim \\ \mathrm{Gal}_K & \longrightarrow & \mathrm{Gal}_k \end{array} \quad (3)$$

we get s is a section of the exact sequence 1. Hence it suffices to prove the splitting property for K' . Set Σ to be the set of such extension K'/K . Obviously, $K \in \Sigma$ so that Σ is not empty and for any chain $\{K'_\alpha\}$, take $K'' = K(\cup_\alpha K'_\alpha)$. It is clear that K'' is algebraic over K and k'' is purely inseparable over k . Then by Zorn's Lemma, we can choose K'/K maximal in Σ .

Now replace K by K' . By our choice, we immediately get K is perfect. Consider the value group $\Gamma := \{v(a) \mid a \in K^\times\}$, where v is the valuation on K . Claim that Γ is divisible and we will prove it later. Let $\mathrm{Char} k = p$. Then the value group of the maximal unramified extension of K is still divisible so that there is no nontrivial tame extension over K^{un} . Therefore the tame inertia part vanishes and I_K equal to the wild inertia part, which is a profinite p -group.

When $p = 0$, the wild inertia part also vanishes so that $I_K = 0$ and there is nothing to prove. When $p \neq 0$ and $I_K \neq 0$, consider the Frattini subgroup $\Phi(I_K)$ of I_K . Then $I_K/\Phi(I_K) \neq 0$ and there is an exact sequence

$$0 \longrightarrow I_K/\Phi(I_K) \longrightarrow \mathrm{Gal}_K/\Phi(I_K) \longrightarrow \mathrm{Gal}_k \longrightarrow 0 \quad (4)$$

By Artin-Schreier theory, we get $H^2(\mathrm{Gal}_k, I_K/\Phi(I_K)) = 0$ and hence the exact sequence 4 splits. Take a section $s : \mathrm{Gal}_k \rightarrow \mathrm{Gal}_K/\Phi(I_K)$ and define H to be the subgroup of Gal_K corresponding to $\mathrm{im} s$. Clearly, H is a closed subgroup. Set $L = (K^s)^{\Phi(I_K)}$ and $K' = (K^s)^H$ with residue field l and k' respectively. Want to show k'/k is purely inseparable.

In fact, as $\mathrm{Gal}(L/K') = H/\Phi(I_K) \cong \mathrm{im} s \cong \mathrm{Gal}_k$, we get a Gal_k -action on l with fixed locus k'' purely inseparable over k' . While $(k^s)^{\mathrm{Gal}_k} = k$, there must be no separable

intermediate field of k''/k . Thus k'/k is purely inseparable and $K \subsetneq K'$, contradicting to maximality of K . Conclude that $I_K = 0$ and hence the exact sequence 1 splits. $\square$

Remark 1. To see $H^2(\text{Gal}_k, I_K/\Phi(I_K)) = 0$, one may replace $I_K/\Phi(I_K)$ by direct limit of finite dimensional $\mathbb{F}_p$ -vector subspaces V_α contained in it. While $H^2(\text{Gal}_k, \mathbb{F}_p) = 0$, we then get $H^2(\text{Gal}_k, I_K/\Phi(I_K)) = \varinjlim_\alpha H^2(\text{Gal}_k, V_\alpha) = 0$.

Lemma 2. Assume K has the maximal property as above. Then Γ is divisible.

Proof. Suppose not, then there exists some positive number n such that $\Gamma \xrightarrow{\times n} \Gamma$ is not surjective. We may choose n minimal and take $a \in K^\times$ such that $v(a) \notin n \cdot \Gamma$. Clearly, $v(a) \neq 0$ and we may assume $v(a) > 0$. In addition, we should have n prime, otherwise for nontrivial factor m of n , $m \cdot \Gamma = \Gamma$ and hence there exists $b \in K$ such that $v(b) = \frac{v(a)}{m} = \frac{\frac{n}{m} \cdot v(a)}{n}$, contradicting to $\frac{n}{m} \cdot \Gamma = \Gamma$.

Consider the extension $L := K[x]/(x^n - a)$ over K , which is separable and hence $L = K(\alpha)$ for some root α of $x^n - a$. Want to show extension of residue field l/k is purely inseparable. Suppose there is some intermediate field $l' \neq k$ separable over k . Then by Primitive Root Theorem, there exists $\beta \in L \setminus K$ such that $k(\bar{\beta}) = l'$. We can write $\beta = a_m \alpha^m + a_{m-1} \alpha^{m-1} + \dots + a_0$, where $a_i \in K$ and $a_m \neq 0$. And we may assume $m < n$ is minimal.

Clearly, by definition, we firstly have $v(a_i \alpha^i) \neq 0$ for all $i > 0$. If $v(a_i \alpha^i) > 0$ for some $i > 0$, then by ultrametric inequality, we must have $v(\beta - a_i \alpha^i) = 0$ and we can replace β by $\beta - a_i \alpha^i$. So we may assume $v(a_i \alpha^i) < 0$ for all $i > 0$.

Note that for all $0 < i < j < n$ with $a_i, a_j \neq 0$, we must have $v(a_i \alpha^i) \neq v(a_j \alpha^j)$, otherwise $v(a_i \cdot a_j^{-1}) = \frac{(i-j) \cdot v(a)}{n}$, contradiction! Hence by valuation property, we get $v(a_m \alpha^m + a_{m-1} \alpha^{m-1} + \dots + a_1 \alpha) = v(a_i \alpha^i)$ for some i . While $v(\beta) = 0$, we must get $v(a_0) = v(a_i \alpha^i)$, contradiction! $\square$

Remark 2. The reduction of n to a prime number is necessary, otherwise the residue field extension l/k might be not purely inseparable. Consider the following example.

Let $K = \mathbb{Q}((t))$ be the field of Laurent series over $\mathbb{Q}$ equipped with the t -adic valuation $v(t) = 1$. Then K is a henselian perfect valued field with value group $\Gamma = \mathbb{Z}$ and residue field $k = \mathbb{Q}$.

Take $a = 2t^2$. Then $v(a) = 2$ and $\frac{v(a)}{6} = \frac{1}{3} \notin \Gamma$. Consider $L := K[x]/(x^6 - a)$. Let α be a root of $x^6 - a$. Since $v(\alpha) = \frac{1}{3}$, the extension has enlarged the value group as expected, while the residue field extension l/k is not purely inseparable.

In fact, if take $\beta = \frac{\alpha^3}{t}$ of valuation 0. then it gives a residue class $\bar{\beta} \in l$. Moreover,

$$\beta^2 = \frac{\alpha^6}{t^2} = \frac{2t^2}{t^2} = 2. \quad (5)$$

Thus $\bar{\beta}$ satisfies $X^2 - 2 \in \mathbb{Q}[X]$. And for any $b \in k = \mathbb{Q}$, note that $v(\beta - b) + v(\beta + b) = v(\beta^2 - b^2) = v(2 - b^2) = 0$, while $v(\beta \pm b) \geq 0$, we get $v(\beta - b) = v(\beta + b) = 0$ so that $\bar{\beta} \notin k$. Hence $X^2 - 2$ is separable and irreducible over k so that

$$\mathbb{Q}(\bar{\beta})/\mathbb{Q} \quad (6)$$

is a nontrivial separable extension. Therefore l/k contains a nontrivial separable subextension and hence is not purely inseparable.

2 Generalization of universal divisibility bound thm

Given a field K with discrete valuation v of characteristic 0 (temporarily preserving char 0 assumption until we check for every detail). Denote K_v is the completion of K with respect to v and I_v the inertia group of K_v^s/K_v , which is canonically isomorphic to the Galois group of K_v^s over K_v^{un} , where K_v^s is the separable closure of K_v with a fixed valuation $\bar{v}$ extending v and K_v^{un} is the maximal unramified extension of K_v .

Definition 1. Let A be an abelian variety over K . For field extension L/K with extending discrete valuation w , we say A has semi-abelian reduction at L if the identity component of special fiber of the Neron model of $A \times_K L$ over valuation ring $\mathcal{O}_w$ is semi-abelian, i.e. an extension

$$0 \longrightarrow T \longrightarrow \mathcal{A}_l^0 \longrightarrow B \longrightarrow 0 \quad (7)$$

where l is the residue field of $\mathcal{O}_w$, T an affine torus and B an abelian variety.

In [3, see p.354-355], Grothendieck defined an open subgroup $I_{A,v}$ of I_v normal in Gal_{K_v} such that for any finite separable extension L/K with valuation $w = \bar{v}|_L$, A has semi-abelian reduction at L if and only if I_w is a subgroup of $I_{A,v}$. In particular, $I_{A,v}$ cuts out the smallest Galois extension of K_v^{un} over which A has semi-abelian reduction. Denote $\Phi_{A,v} := I_v/I_{A,v}$ and $(K_v^{\text{un}})_A = (K_v^s)^{I_{A,v}}$.

To study semi-abelian reduction, it is natural to research on $\Phi_{A,v}$. In fact, Silverberg and Zarhin proved in their paper [4] that for any such pair (K, v, A) with $\dim A = g$, we have $\#\Phi_{A,v} \mid M(2g)$, where $M(2g)$ is the Minkovski bound. And if define $d_v(A)$ as $\min\{[L : K] \mid A \text{ has semi-abelian reduction at } L\}$, then a next question is that can we show $d_v(A) \mid M(2g)$ for any such pair (K, v, A) with $\dim A = g$?

This question has been answered for abelian varieties over number field by Séverin Philip in his series of papers, in which case instead of considering just one place, we seek for semi-abelian reduction everywhere and $d_v(A)$ would be replaced by $d(A)$. [5] and [6] formulate the universal divisibility bound theorem in the number field case with construction A with $d(A) = M(2g) \cdot 2^{1-g}$ for all $g \geq 1$ implying that $M(2g)$ is optimal at odd primes but 2 is problematic. While recently, Philip posts another paper [7] including a new construction at prime 2 so that $M(2g)$ is really the optimal universal divisibility bound. And I think the following lemma, see in [5, Lemma 3.1], is an important basestone of his work.

Lemma 3. Let A be an abelian variety over a local field K_v . Then there is a finite separable extension L_w/K_v such that A has semi-abelian reduction at L_w with $[L_w : K_v] = \#\Phi_{A,v}$. Moreover, for any finite separable extension L_w/K_v such that A has semi-abelian reduction at L_w , we have that $\#\Phi_{A,v} \mid [L_w : K_v]$.

Remark 3. For the proof of the first part, the local field assumption implies that $\text{Gal}_{k_v} \cong \widehat{\mathbb{Z}}$. Since $\widehat{\mathbb{Z}}$ is projective, the following exact sequence splits

$$0 \longrightarrow \Phi_{A,v} \longrightarrow \text{Gal}((K_v^{\text{un}})_A/K_v) \longrightarrow \text{Gal}_{k_v} \longrightarrow 0 \quad (8)$$

so that there is a section $s : \text{Gal}_{k_v} \rightarrow \text{Gal}((K_v^{\text{un}})_A/K_v)$ whose image is closed and cuts out a finite separable extension L_w/K_v . In particular, we have $[L_w : K_v] = \sharp\Phi_{A,v}$ and $L_w \cdot K_v^{\text{un}} = (K_v^{\text{un}})_A$. Take the maximal unramified extension of L_w in K_v^s , denoted by L_w^{un} . Then $L_w^{\text{un}} \supseteq (K_v^{\text{un}})_A$ and hence A has semi-abelian reduction at L_w .

So if we want to extend the universal divisibility bound theorem to general case, it is natural to question can we get splitting exact sequence as Lemma 3. And this is where Gabber's observation comes in.

Corollary 1. Notations as the beginning of this section. There is a finite separable extension L/K such that A has semi-abelian reduction at L with $[L : K] = \sharp\Phi_{A,v}$. Moreover, for any finite separable extension L/K such that A has semi-abelian reduction at L , we have that $\sharp\Phi_{A,v} \leq [L : K]$.

Proof. Since K is discretely valued, K_v is henselian. Then by Lemma 1, we get a splitting exact sequence

$$0 \longrightarrow I_v \longrightarrow \text{Gal}_{K_v} \longrightarrow \text{Gal}_{k_v} \longrightarrow 0 \quad (9)$$

Assume $s : \text{Gal}_{k_v} \rightarrow \text{Gal}_{K_v}$ is a section. Then for any normal subgroup $I' \trianglelefteq \text{Gal}_{K_v}$ contained in I_v , it is clear that the composition $\text{Gal}_{k_v} \xrightarrow{s} \text{Gal}_{K_v} \rightarrow \text{Gal}_{K_v}/I' \rightarrow \text{Gal}_{k_v}$ is still the identity map. Thus the following exact sequence splits

$$0 \longrightarrow I_v/I' \longrightarrow \text{Gal}_{K_v}/I' \longrightarrow \text{Gal}_{k_v} \longrightarrow 0 \quad (10)$$

Take I' to be $I_{A,v}$ and we get splitting exact sequence

$$0 \longrightarrow \Phi_{A,v} \longrightarrow \text{Gal}((K_v^{\text{un}})_A/K_v) \longrightarrow \text{Gal}_{k_v} \longrightarrow 0 \quad (11)$$

Then by same argument as Lemma 3, we get finite separable extension L_w/K_v of degree $\sharp\Phi_{A,v}$ such that A has semi-abelian reduction at L_w .

Take a primitive element of L_w/K_v , say α . Assume $f(x)$ is the minimal polynomial of α over K_v . We may assume that $f(x) \in \widehat{\mathcal{O}_v}[x]$, where $\mathcal{O}_v$ is the valuation ring of K and $\widehat{\mathcal{O}_v}$ is the completion.

Then we can pick $f_n(x) \in \mathcal{O}_v[x]$ such that $\deg f_n(x) = \deg f(x)$ and all coefficients of $f(x) - f_n(x)$ are of valuation greater than or equal to n . And Krasner's Lemma tells us that for n large enough, we can find a root β of $f_n(x)$ such that $K_v(\beta) = K_v(\alpha) = L_w$.

Claim that $L := K(\beta)$ with valuation $w|_L$ is our desired field. As

$$\begin{aligned} [L_w : K_v] &= \deg f(x) \\ &\geq [L : K] \\ &\geq [K_v(\beta) : K_v] \\ &= [L_w : K_v] \end{aligned} \quad (12)$$

we get $[L : K] = \sharp\Phi_{A,v}$. Notice that $L_w^{\text{un}} \supseteq (K_v^{\text{un}})_A$ and hence A has semi-abelian reduction at L .

For the second part, given finite separable extension L/K such that A has semi-reduction at L , we have that $L_w^{\text{un}} \supseteq (K_v^{\text{un}})_A$ and hence $\sharp\Phi_{A,v} \mid [I_v : I_w] = [L_w^{\text{un}} : K_v^{\text{un}}]$. Since K_v and L_w are both henselian, there is one-to-one correspondence between finite separable extension of residue field and unramified extension. While the residue field of $L_w \cdot K_v^{\text{un}}$ is $l \cdot k^s = l^s$, we then get $L_w \cdot K_v^{\text{un}} = L_w^{\text{un}}$. Hence

$$[L_w^{\text{un}} : K_v^{\text{un}}] \leq [L_w : K_v] \leq [L : K] \quad (13)$$

Conclude that $\sharp\Phi_{A,v} \leq [L : K]$. $\square$

Remark 4. Originally, I want a result like $\sharp\Phi_{A,v} \mid [L : K]$. But there are finitely many extensions of v on L , and we have an equation that $\sum_i e_i f_i = [L : K]$, which does not guarantee $e_i \mid [L : K]$ for all i .

Corollary 2. As a corollary, $d_v(A) = \sharp\Phi_{A,v}$.

By [5] and [4], we then conclude that $d_g := \text{lcm}\{d_v(A) \mid \dim A = g\}$ is bounded by the Minkovski bound $M(2g)$. More precisely, their results tell us that $d_v(A) \mid M(2g)$ for any A of dimension g . And the following is the target theorem which I want to establish.

Theorem 1 (Target Theorem). *Notations as above, $M(2g)$ is the optimal universal divisibility bound i.e. $M(2g) = \text{lcm}\{d_v(A) \mid \forall (K, v, A) \text{ with } \dim A = g\}$.*

Reason. By Corollary 2, $M(2g)$ is a universal divisibility bound. And combining results from papers [5], [6] and [7], we conclude that $M(2g)$ is optimal. $\square$

References

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